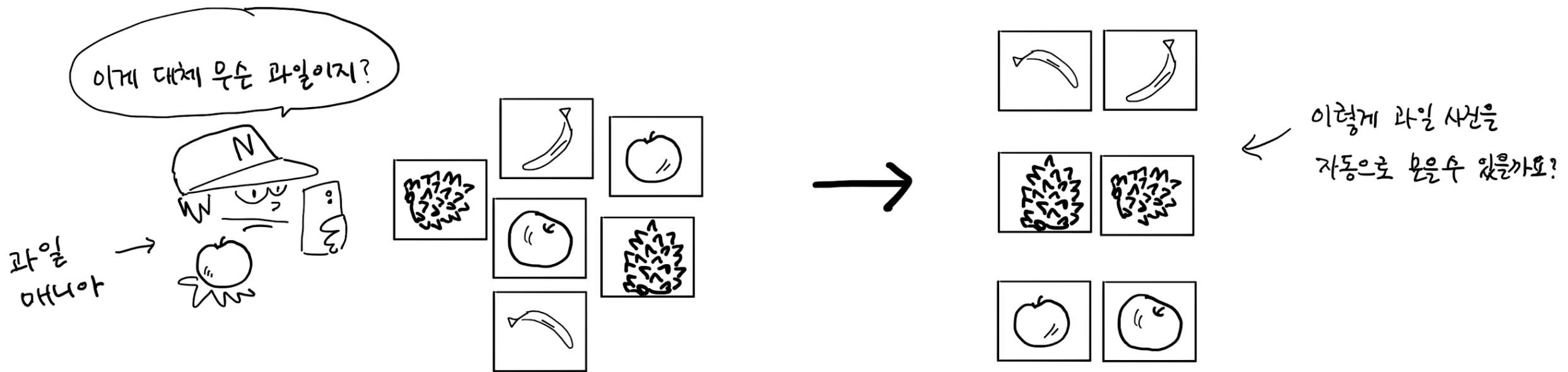


· 1. 비지도학습 (군집 알고리즘)

비지도 학습

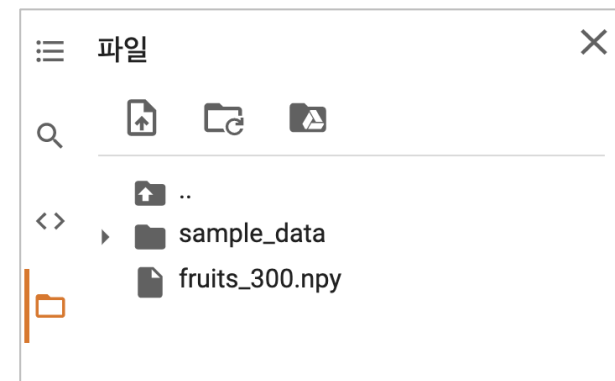
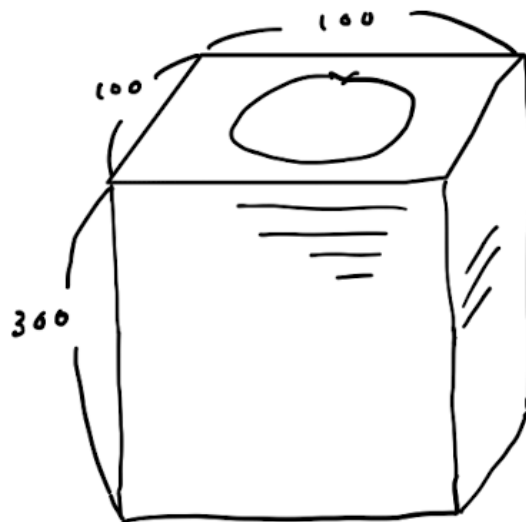


과일 데이터 준비하기

```
!wget https://bit.ly/fruits_300_data -O fruits_300.npy
```

```
import numpy as np  
import matplotlib.pyplot as plt
```

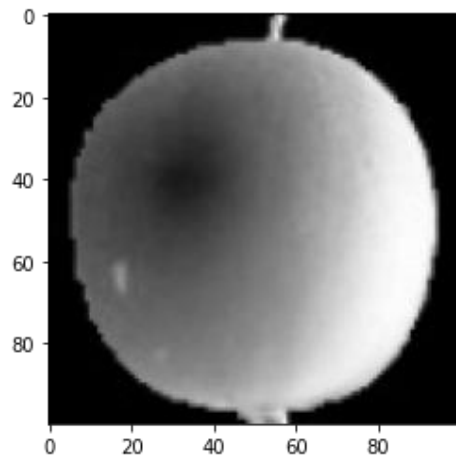
```
fruits = np.load('fruits_300.npy')  
print(fruits.shape)  
(300, 100, 100)
```



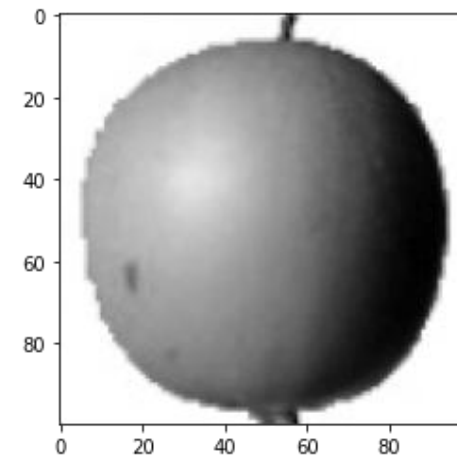
샘플 확인

```
print(fruits[0, 0, :])  
[ 1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  2  1  
 2  2  2  2  2  2  1  1  1  1  1  1  1  1  2  3  2  1  
 2  1  1  1  1  2  1  3  2  1  3  1  4  1  2  5  5  5  
19 148 192 117 28  1  1  2  1  4  1  1  3  1  1  1  1  1  
 2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  
 1  1  1  1  1  1  1  1  1  1  1]
```

```
plt.imshow(fruits[0], cmap='gray')  
plt.show()
```



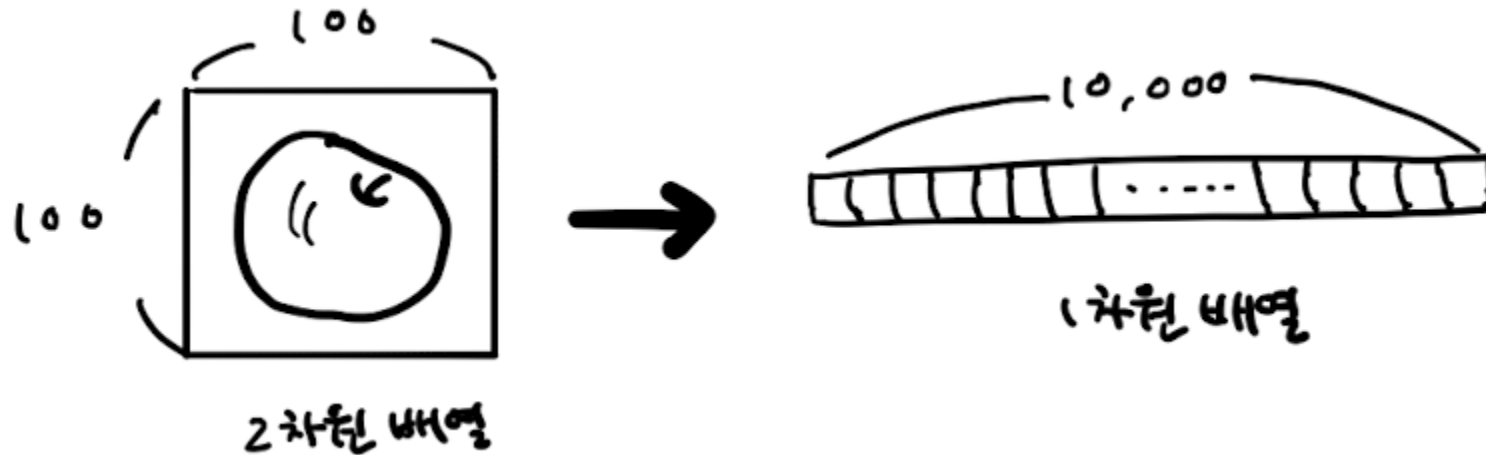
```
plt.imshow(fruits[0], cmap='gray_r')  
plt.show()
```



샘플 차원 변경하기

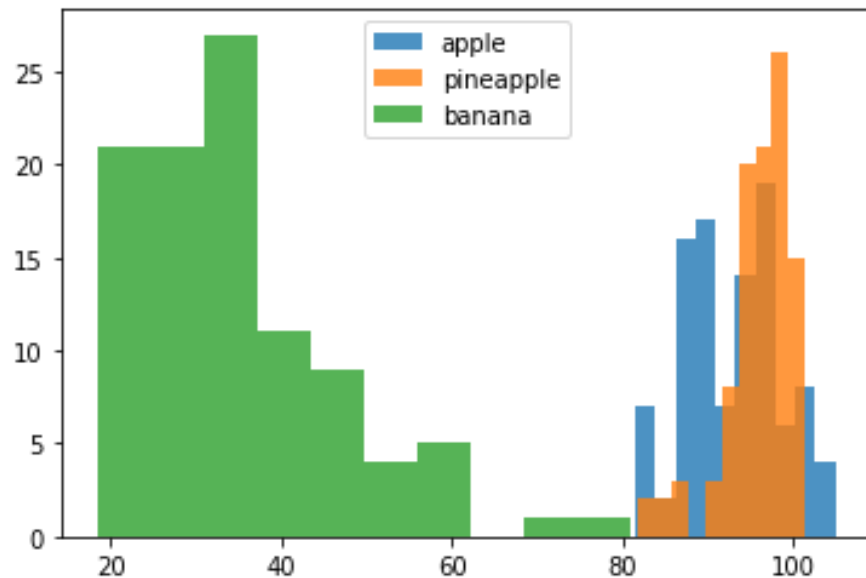
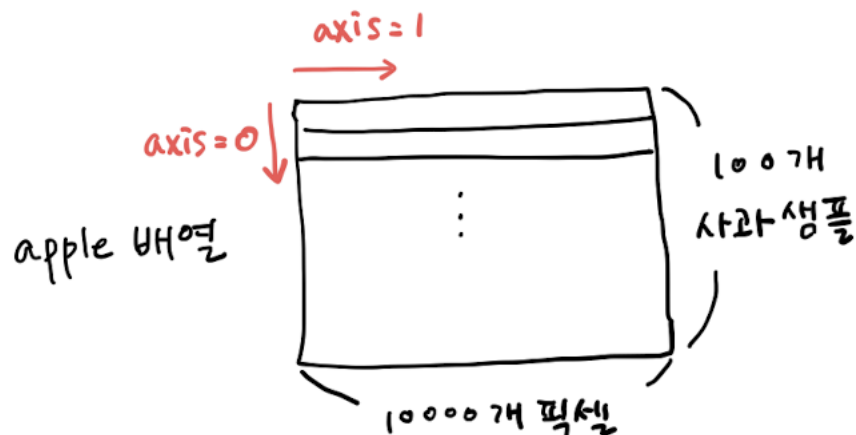
```
apple = fruits[0:100].reshape(-1, 100*100)
pineapple = fruits[100:200].reshape(-1, 100*100)
banana = fruits[200:300].reshape(-1, 100*100)
```

```
print(apple.shape)
(100, 10000)
```



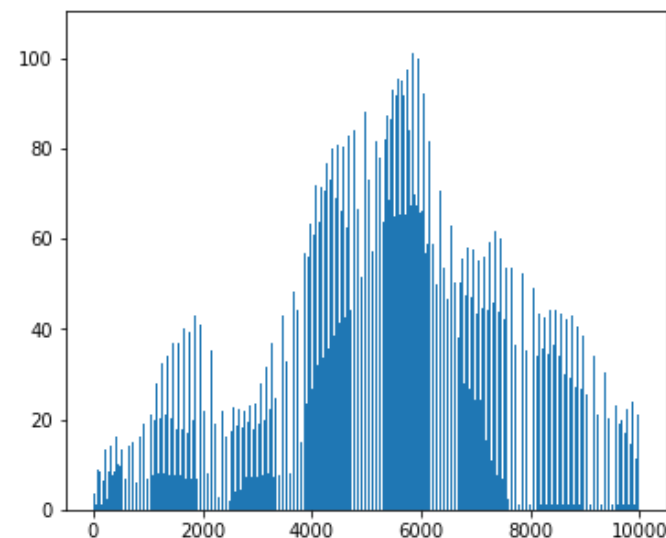
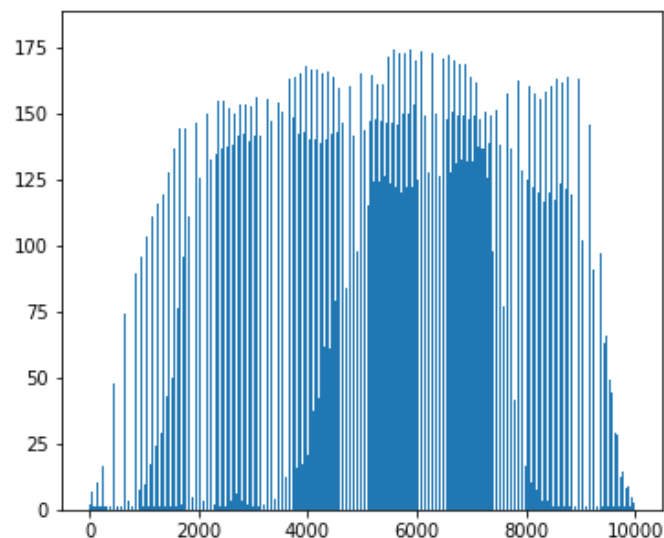
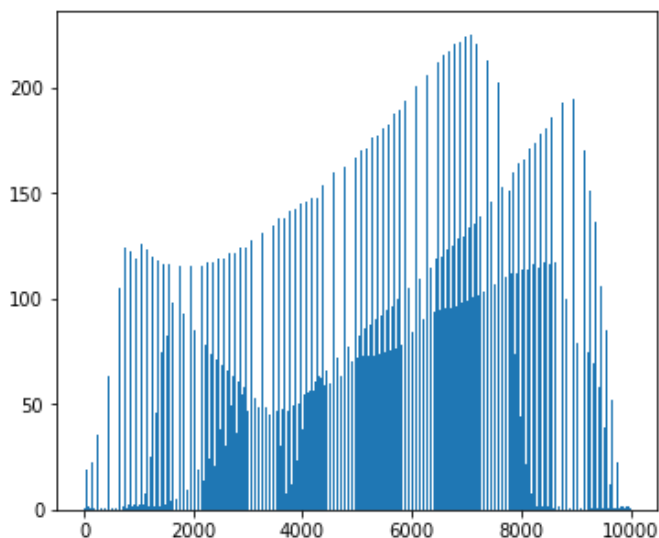
각 샘플 평균의 히스토그램

```
plt.hist(np.mean(apple, axis=1), alpha=0.8)
plt.hist(np.mean(pineapple, axis=1), alpha=0.8)
plt.hist(np.mean(banana, axis=1), alpha=0.8)
plt.legend(['apple', 'pineapple', 'banana'])
plt.show()
```



픽셀 평균의 막대그래프

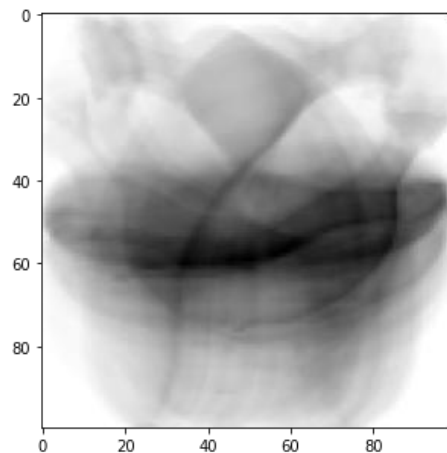
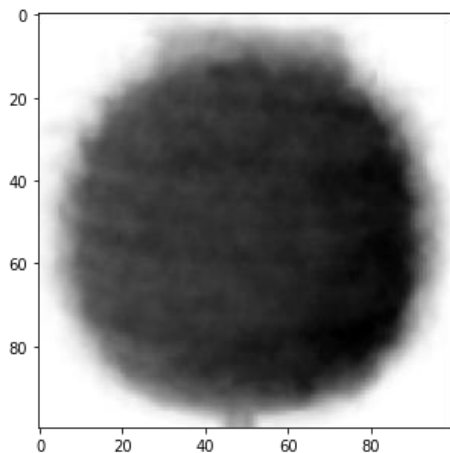
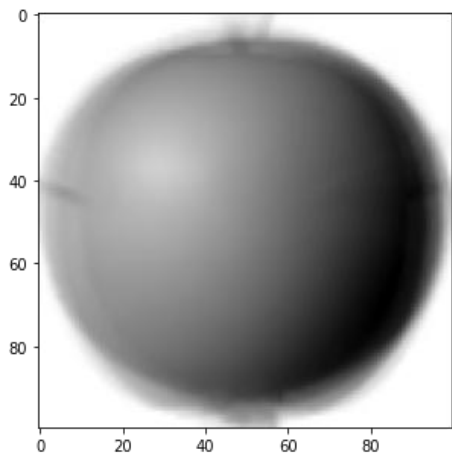
```
fig, axs = plt.subplots(1, 3, figsize=(20, 5))
axs[0].bar(range(10000), np.mean(apple, axis=0))
axs[1].bar(range(10000), np.mean(pineapple, axis=0))
axs[2].bar(range(10000), np.mean(banana, axis=0))
plt.show()
```



평균 이미지 그리기

```
apple_mean = np.mean(apple, axis=0).reshape(100, 100)
pineapple_mean = np.mean(pineapple, axis=0).reshape(100, 100)
banana_mean = np.mean(banana, axis=0).reshape(100, 100)
```

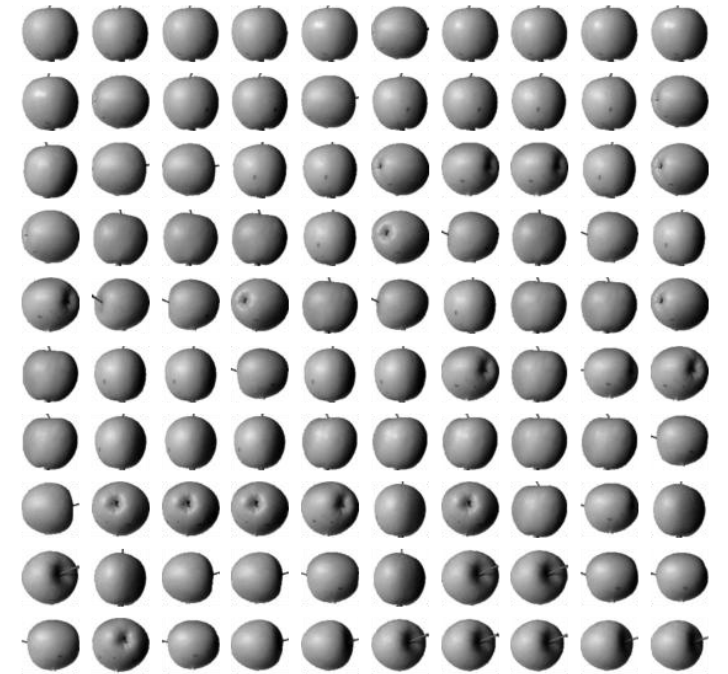
```
fig, axs = plt.subplots(1, 3, figsize=(20, 5))
axs[0].imshow(apple_mean, cmap='gray_r')
axs[1].imshow(pineapple_mean, cmap='gray_r')
axs[2].imshow(banana_mean, cmap='gray_r')
plt.show()
```



평균과 가까운 사진 고르기

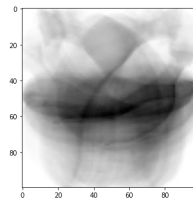
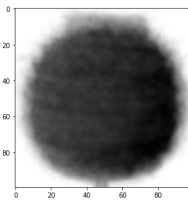
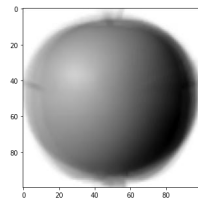
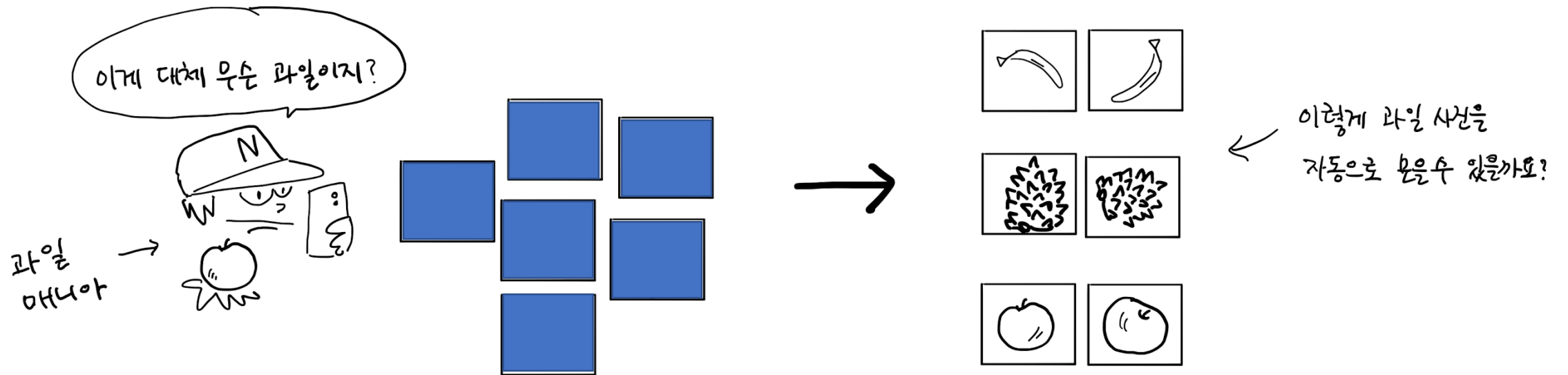
```
abs_diff = np.abs(fruits - apple_mean)
abs_mean = np.mean(abs_diff, axis=(1,2))
print(abs_mean.shape)
(300,)

apple_index = np.argsort(abs_mean)[:100]
fig, axs = plt.subplots(10, 10, figsize=(10,10))
for i in range(10):
    for j in range(10):
        axs[i, j].imshow(fruits[apple_index[i*10 + j]], cmap='gray_r')
        axs[i, j].axis('off')
plt.show()
```

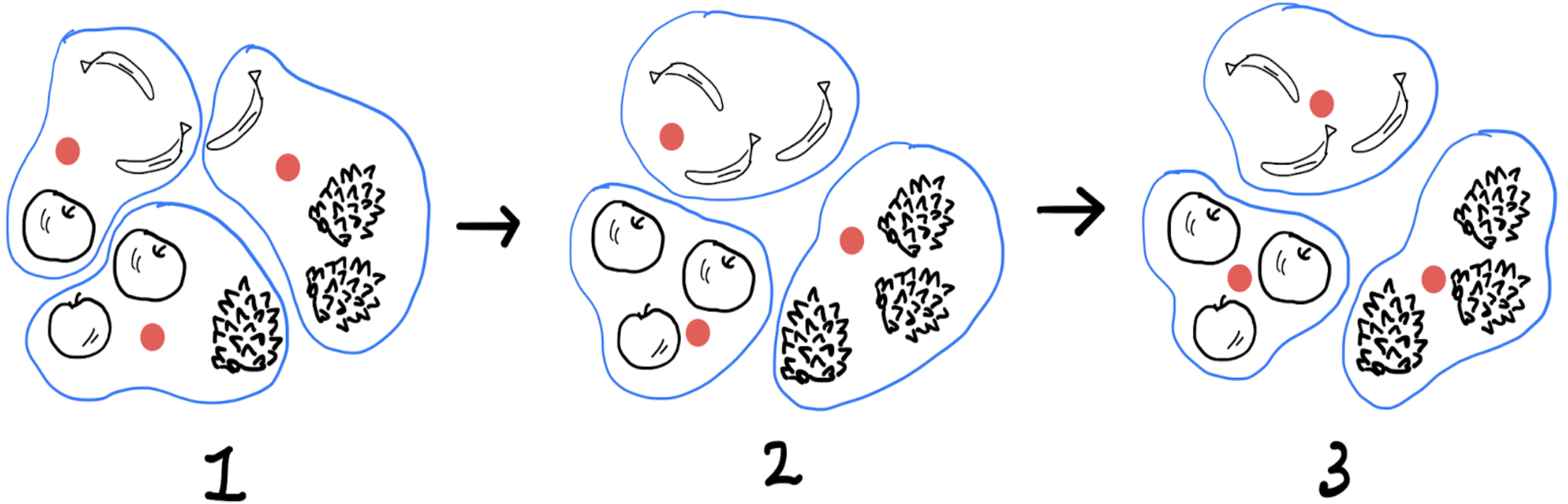


• 2. k-mean 알고리즘

군집



k-mean 알고리즘



k-mean 알고리즘 - 모델 훈련

```
from sklearn.cluster import KMeans
```

```
km = KMeans(n_clusters=3, random_state=42)  
km.fit(fruits_2d)
```

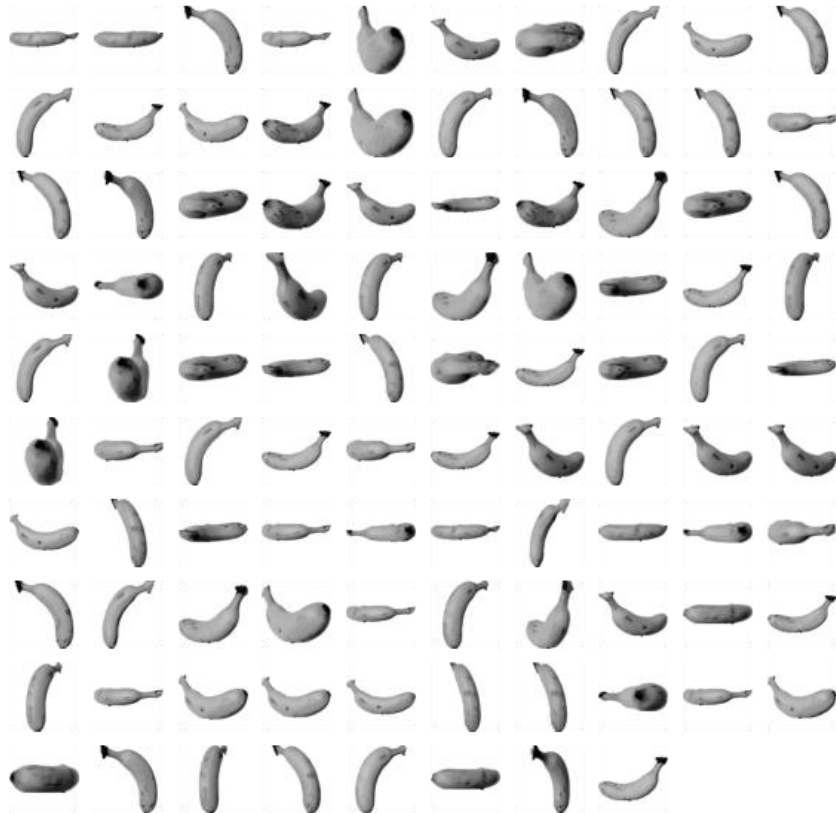
```
print(km.labels_)
```

```
[0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
 0 0 0 0 0 2 0 2 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 0 2 2 0 0 0 0 0 0 2 0  
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 2 2 2 2 2 2 2 2 2  
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2  
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2  
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
 1 1 1 1]
```

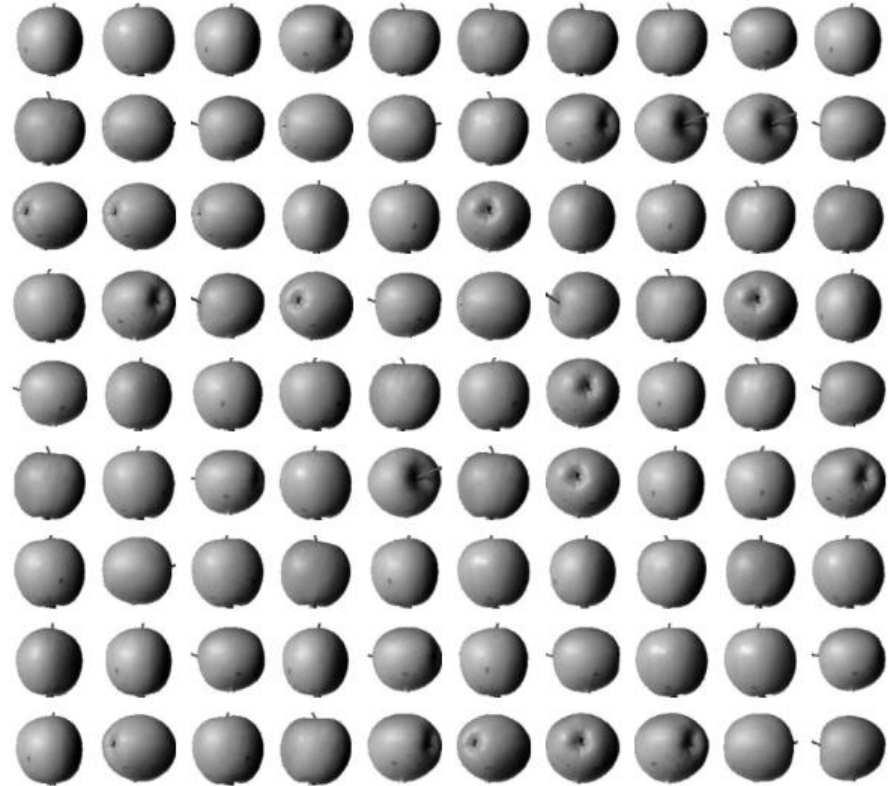
```
print(np.unique(km.labels_, return_counts=True))
```

클러스터 출력

```
draw_fruits(fruits[km.labels_==1])
```

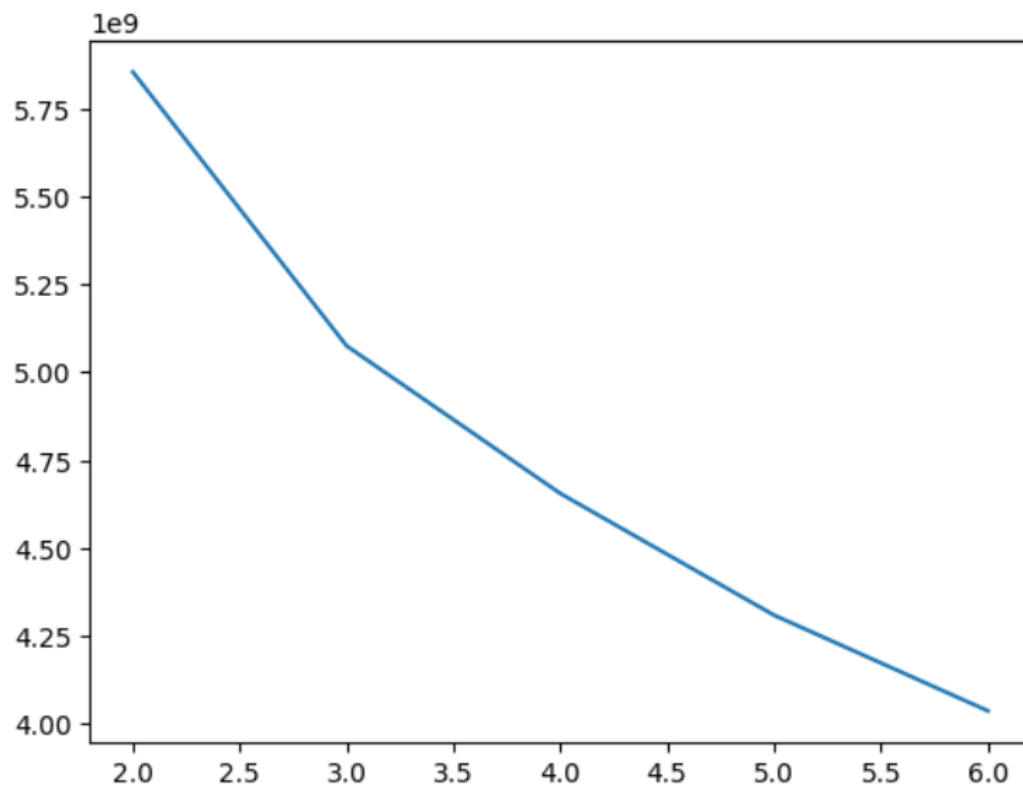


```
draw_fruits(fruits[km.labels_==2])
```



최적의 k 찾기

```
inertia = []  
for k in range(2, 7): # 임의로 2~6까지 지정해 보자  
    km = KMeans(n_clusters=k, random_state=42)  
    km.fit(fruits_2d)  
    inertia.append(km.inertia_)  
  
plt.plot(range(2, 7), inertia)  
plt.show()
```



확인 문제

1. 비지도 학습(Unsupervised Learning)에 대한 설명으로 가장 옳은 것은?

- ① 정답(label)이 있는 데이터를 이용해 학습한다
 - ② 입력 데이터와 출력 데이터 모두 필요하다
 - ③ 정답 없이 데이터의 구조나 패턴을 찾는다
 - ④ 반드시 회귀 문제에만 사용된다
-

2. 다음 중 비지도 학습의 대표적인 알고리즘은?

- ① K-Nearest Neighbors
- ② K-Means Clustering
- ③ Linear Regression
- ④ Logistic Regression

확인 문제

3. K-Means에서 inertia의 의미로 가장 적절한 것은?

- ① 클러스터 개수
 - ② 중심점의 위치
 - ③ 데이터 간 거리의 평균
 - ④ 클러스터 내 거리 제곱합
-

4. K-Means 알고리즘의 핵심 단계가 아닌 것은?

- ① 초기 중심 설정
- ② 데이터 할당
- ③ 중심 업데이트
- ④ 역전파 수행

확인 문제

5. K-Means에서 centroid는 무엇인가?

- ① 가장 먼 데이터
 - ② 클러스터 중심 평균값
 - ③ 랜덤 데이터
 - ④ 정답 레이블
-

6. Elbow graph에서 좋은 K 값은?

- ① inertia가 가장 큰 지점
- ② 그래프가 급격히 꺾이는 지점
- ③ $K=1$
- ④ 무조건 $K=10$

